

MEMORANDUM

Network Operations Strategy — ETA Optimization & Bottleneck Remediation

1. Executive Summary

Delivery ETA accuracy is one of the most visible drivers of customer trust and operational cost at Delhivery. Every percentage point of ETA error translates into missed delivery promises, escalations, and reactive resourcing across the network. Our review of 144,867 trip segments across 1,657 facilities and 2,783 active corridors found that the current routing-engine (OSRM) estimate is wrong by more than 15% on the vast majority of deliveries — only 3.8% of OSRM-based ETAs land within 15% of the actual delivery time, and the average actual transit time runs at roughly 2.25x the OSRM estimate.

This gap is not random. It is concentrated in a small number of structurally critical hubs and a defined set of high-delay corridors — meaning it is addressable through targeted operational intervention rather than network-wide investment.

To address this, we introduced graph network intelligence: modeling Delhivery's facility network as a directed graph of corridors, computing structural-importance metrics (centrality, PageRank) for every facility, and using graph embeddings (Node2Vec) to give our ETA prediction model awareness of each facility's position in the network — not just its raw distance and traffic figures.

Key Outcomes Achieved

- **Improved ETA accuracy:** the graph-enhanced model reduced mean absolute ETA error from 34.2 minutes to 27.7 minutes — a 19.1% reduction — and raised the share of deliveries with ETAs within 15% of actual time from 56.1% to 61.3% (+5.2 percentage points).
- **Identified the network's critical bottlenecks:** a ranked list of the top facilities whose congestion and delay patterns disproportionately drive systemic SLA breaches.
- **Quantified a clear remediation roadmap:** upgrading the top three bottleneck hubs alone is projected to reduce SLA breaches by ~3.0% and recover an estimated ₹196,350 in revenue at risk, based on illustrative cost assumptions that should be recalibrated with actual Delhivery figures.
- **Surfaced a route-type optimization opportunity:** 14 corridors are currently using a suboptimal FTL/Carting assignment, representing an estimated 53 hours of recoverable transit time.

2. Key Findings

Top 5 Structural Bottleneck Hubs

Using a composite Structural Risk Score (combining network centrality, PageRank influence, average outbound delay, and traffic volume), five facilities stand out as carrying outsized structural risk for the entire network:

Rank	Facility	Structural Risk Score	Betweenness Centrality	SLA Breach Contribution
1	Gurgaon Bilaspur HB (Haryana)	0.807	0.330	25.9%
2	Bangalore Nelamangala H (Karnataka)	0.484	0.187	14.6%
3	Hyderabad Shamshabad H (Telangana)	0.342	0.139	6.1%
4	Kolkata Dankuni HB (West Bengal)	0.303	0.136	5.3%

5	Bhiwandi Mankoli HB (Maharashtra)	0.300	0.104	10.6%
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These five hubs alone account for over 62% of all SLA-breach segments network-wide, despite representing less than 0.4% of all facilities — a textbook 80/20 concentration that makes them the highest-leverage intervention points in the network.

Structural Risk Observations

- **Network connectivity is fragmented:** the corridor graph splits into 64 separate weakly-connected components. The largest contains 1,353 facilities (81.7% of the network) and 2,463 corridors (88.5%) — this main component is the basis for all centrality and embedding analysis.
- **Betweenness centrality concentrates at a single node:** Gurgaon Bilaspur's betweenness score (0.330) is nearly double the next-highest facility, meaning a disproportionate share of network paths route through it — any disruption here has the widest ripple effect of any facility in the system.
- **High structural importance correlates with high delay exposure:** four of the five top-risk hubs also rank in the top 10 for SLA-breach contribution, confirming that structural centrality and operational delay are reinforcing, not independent, problems.

Major Delayed Corridors

93.4% of all 2,783 corridors (2,598 corridors) run at a median delay of more than 20% over the OSRM estimate — confirming that delay is the network norm, not the exception, for OSRM-based planning. The corridors with the single largest cumulative time impact are:

Corridor	Median Delay Ratio	Trip Volume	Total Delay Impact (minutes)
Shirpur (Maharashtra) -> Dhule MIDC (Maharashtra)	21.2x	14	9,338
Hazrat Jandaha (Bihar) -> Muzaffarpur Bbganj (Bihar)	11.4x	14	6,893
Benipatti (Bihar) -> Muzaffarpur Bbganj (Bihar)	10.0x	15	6,789
Simlapal (West Bengal) -> Hura Central (West Bengal)	14.2x	8	5,502
Helencha (West Bengal) -> Kolkata Dankuni HB (West Bengal)	15.4x	5	5,299

These corridors show extreme delay ratios (10x–21x OSRM estimates) but relatively low trip volumes — they represent route-planning failures (likely poor OSRM road-network mapping for regional/rural routes) rather than capacity bottlenecks, and are addressed differently from the high-volume hub bottlenecks above.

Key Causes of ETA Deviation

- **OSRM underestimates real-world transit time across the board:** average delay ratio is 2.25x for Carting and lower (but still substantial) for FTL — OSRM's free-flow routing assumptions do not reflect actual road conditions, loading/unloading time, or hub dwell time.
- **Carting routes run hotter than FTL:** Carting shows a higher average delay ratio (2.67x vs 2.06x) and a marginally lower SLA breach rate (85.3% vs 91.1%) than FTL, but FTL shows a higher average delay in absolute minutes (19.4 vs 14.0) — both route types contribute meaningfully to the problem, for different reasons.

- **Network position matters beyond distance and traffic:** graph embeddings — capturing each facility's position in the corridor network — contributed 7.2% of total predictive signal in the enhanced model, on top of conventional features like OSRM time/distance, time-of-day, and route type.

3. Model Impact

We benchmarked three approaches to ETA prediction against the existing OSRM estimate:

Model	MAE (min)	RMSE (min)	R ²	% Within 15% of Actual ETA
OSRM Estimate (current state)	204.0	366.9	0.624	3.8%
Random Forest (baseline ML)	35.4	69.0	0.987	56.9%
XGBoost (baseline ML)	34.2	63.1	0.989	56.1%
XGBoost + Graph Embeddings (recommended)	27.7	50.9	0.993	61.3%

Translating the Numbers into Business Value

- **From OSRM to Machine Learning:** moving from the current OSRM-based estimate to a standard ML model already cuts average ETA error from over 3 hours to roughly 34 minutes, and increases the share of "accurate" ETAs (within 15% of actual) from under 4% to over 56% — a transformational jump in baseline reliability.
- **From Baseline ML to Graph-Enhanced ML:** adding network-structure intelligence on top of the baseline model reduces average ETA error by a further 6.5 minutes per delivery (19.1% improvement) and pushes "accurate" ETAs from 56.1% to 61.3% — an additional 5.2 percentage points of deliveries where the customer-facing promise will be met without ML-side adjustment.
- **What this means operationally:** more accurate ETAs reduce the volume of customer escalations, support-desk inquiries, and reactive re-planning by ops teams, while improving the credibility of delivery-time commitments used in customer communications and SLA contracts.

We recommend the graph-enhanced model (XGBoost + Node2Vec) as the production ETA engine, given it outperforms all alternatives on every metric without requiring a fundamentally different operating model — it is a drop-in upgrade to the existing ML-based ETA pipeline.

4. Operational Recommendations — Bottleneck Hubs

For each of the top structural bottleneck hubs, we outline why it matters, the operational risk it represents, and the recommended intervention.

1. Gurgaon Bilaspur HB (Haryana) — Highest Priority

- **Why it matters:** highest structural risk score (0.807) in the network by a wide margin, with the highest betweenness centrality (0.330) — nearly double any other facility — and the single largest SLA-breach contribution (25.9%).
- **Operational risk:** as the most heavily-traversed node in the network, any congestion or delay here propagates to the largest number of downstream corridors. This is a single point of failure for North Indian network throughput.
- **Recommended intervention:** priority capacity expansion (additional sorting lines / dock doors) and parallel-routing analysis to identify alternate paths that can be activated during peak congestion, reducing reliance on this single node for cross-regional transfers.

2. Bangalore Nelamangala H (Karnataka)

- **Why it matters:** second-highest structural risk score (0.484), second-highest betweenness centrality (0.187), and the second-largest SLA-breach contribution (14.6%) — a critical hub for South Indian network flows.
- **Operational risk:** the corridor audit shows this facility's outbound Carting corridors to Bengaluru Bommasandra are among the highest-volume corridors in the network and also exceed the 20% delay threshold — meaning high-volume traffic is routinely running late through this hub.
- **Recommended intervention:** process redesign at the Carting handover point with Bengaluru Bommasandra HB — likely candidate for staffing improvements during the identified peak windows, and a review of whether some Carting volume should shift to FTL (see Section 6).

3. Hyderabad Shamshabad H (Telangana)

- **Why it matters:** third-highest structural risk score (0.342) with significant betweenness centrality (0.139), indicating an important role connecting Telangana to the broader network.
- **Operational risk:** moderate SLA-breach contribution (6.1%) relative to its centrality suggests this hub is structurally important but not yet a major delay source — an opportunity for preventive investment before it becomes a bottleneck as volume grows.
- **Recommended intervention:** capacity planning review ahead of projected volume growth — staffing and throughput audits now are lower-cost than reactive fixes once delay patterns emerge.

4. Kolkata Dankuni HB (West Bengal)

- **Why it matters:** fourth-highest structural risk score (0.303), and the corridor audit independently flags Helencha -> Kolkata Dankuni HB and Ranaghat -> Kolkata Dankuni HB as among the highest total-delay-impact corridors in the entire network.
- **Operational risk:** this hub appears in both the structural-risk ranking and the corridor-delay ranking — a double signal that inbound flows into this facility are a genuine throughput constraint, not just a statistical artifact.
- **Recommended intervention:** inbound capacity and dock-scheduling review specifically for the Eastern-region feeder corridors (Helencha, Ranaghat, Midnapore) that converge on this hub.

5. Bhiwandi Mankoli HB (Maharashtra)

- **Why it matters:** fifth-highest structural risk score (0.300) but the third-highest SLA-breach contribution (10.6%) among the top five — its delay impact is disproportionate to its centrality ranking.
- **Operational risk:** the corridor audit shows extreme delay ratios on Bhiwandi Mankoli HB's Mumbai-area corridors (Mumbai Hub, Mumbai MiraRd, Pune Tathawde) — several exceeding 2x median delay — indicating a regional congestion cluster around Mumbai's logistics corridor, not an isolated facility issue.
- **Recommended intervention:** treat as part of a Mumbai-Pune regional congestion cluster requiring coordinated intervention — staffing improvements at Bhiwandi combined with parallel routing options through Pune Tathawde to relieve pressure during peak periods.

5. Corridor-Level Recommendations

Corridor analysis reveals two distinct categories of delayed routes requiring different remediation approaches.

Category A — High-Volume Hub Corridors (Capacity-Driven Delay)

Corridors such as Bangalore Nelamangala -> Bengaluru Bommasandra (151 trips, 1.42x median delay) and Bhiwandi Mankoli HB -> Mumbai Hub (105 trips, 2.18x median delay) combine high trip volume with sustained delay above the 20% threshold.

- **Proposed action:** throughput audits and staffing/scheduling adjustments at the receiving facility; these corridors are strong candidates for the hub-level interventions in Section 4.
- **Expected operational impact:** because these corridors carry high volume, even modest per-trip improvements compound into significant aggregate time savings — directly supported by the Section 7 simulation showing hub-level upgrades reduce network-wide SLA breaches.

Category B — Extreme-Outlier Regional Corridors (Routing/Mapping-Driven Delay)

Corridors such as Shirpur -> Dhule MIDC (21.2x delay), Hazrat Jandaha -> Muzaffarpur Bbganj (11.4x), and Benipatti -> Muzaffarpur Bbganj (10.0x) show extreme delay ratios despite low trip volumes (8-15 trips each).

- **Proposed action:** these are most likely OSRM road-network data quality issues for rural/regional routes (the routing engine's road-network model does not reflect actual ground travel times) rather than genuine operational failures. Recommend a targeted ground-truth audit: have regional ops teams confirm actual travel times against OSRM estimates for these specific routes and flag for OSRM map-data correction.
- **Expected operational impact:** correcting the underlying OSRM data for these corridors will directly improve the accuracy of customer-facing ETAs on these routes — and, because the ETA model uses OSRM time as a core input feature, will also improve model accuracy network-wide without retraining.

General SLA Monitoring

- **Proposed action:** implement the corridor-monitoring dashboard (built as part of this analysis) to track all 2,598 corridors currently exceeding the 20% delay threshold, with automated flags when new corridors cross this line.
- **Expected operational impact:** shifts the organization from reactive (responding to individual late-delivery complaints) to proactive (identifying systemic corridor degradation before it accumulates into widespread SLA breaches).

6. FTL vs Carting Recommendations

We built a recommendation model that evaluates each corridor's volume, distance, and delay profile to determine the route type (Full Truck Load vs Carting) most likely to minimize delay. The model achieved 99% classification accuracy in distinguishing corridors where FTL outperforms Carting and vice versa.

Where FTL Should Be Prioritized

- **Profile:** longer-distance corridors (model-recommended FTL corridors average 25.6 km vs 11.6 km for Carting) with moderate trip volumes (average 8.2 trips).
- **Specific opportunity:** three of the highest-volume corridors currently run as Carting but are flagged by the model as better suited to FTL: Bangalore Nelamangala -> Bengaluru Bommasandra (127 trips), Bengaluru Bommasandra -> Bangalore Nelamangala (102 trips), and Delhi Gateway HB -> Gurgaon Bilaspur HB (100 trips) — together representing an estimated 2,744 minutes (~46 hours) of the total 53-hour recoverable opportunity.

Where Carting Remains Efficient

- **Profile:** shorter-distance, higher-frequency corridors (model-recommended Carting corridors average 11.6 km, with 1,060 corridors in this category) — Carting's flexibility and lower per-trip overhead make it the right choice for short-haul, high-frequency distribution.
- **Validation:** the overwhelming majority of corridors (2,769 of 2,783, i.e. 99.5%) are already using the model-recommended route type — confirming that Delhivery's existing route-type allocation

logic is largely sound, with the 14 flagged exceptions representing the genuine optimization opportunity.

Business Implications

- **Limited but real upside:** the 14 flagged corridors represent a modest 53-hour cumulative recoverable opportunity — small relative to total network volume, but concentrated in three high-traffic corridors that would benefit most from re-evaluation.
- **Decision framework as an ongoing tool:** rather than a one-time fix, the recommendation model should be re-run quarterly as corridor volumes and distances evolve, ensuring route-type allocation stays optimal as the network grows.

7. Business Impact

Using the assumptions stated below — which should be recalibrated with Delhivery's actual SLA penalty and capacity-investment figures — we simulated the effect of upgrading the top three bottleneck hubs (Gurgaon Bilaspur HB, Bangalore Nelamangala H, Hyderabad Shamshabad H) by reducing their outbound delay ratio by 20%, a conservative capacity-improvement assumption.

Metric	Current State	After Top-3 Hub Upgrade	Improvement
SLA Breaches (segments)	129,351	125,424	-3,927 (-3.0%)
Revenue at Risk (illustrative, ₹50/breach)	₹6,467,550	₹6,271,200	₹196,350 recovered
Segments affected by upgrade	—	36,662 (25.3% of all segments)	—

- **SLA improvement:** a 20% delay reduction at just three facilities — representing a quarter of total network segment volume — reduces total SLA breaches by approximately 3.0%, demonstrating the high leverage of targeted hub investment versus network-wide initiatives.
- **Revenue-at-risk recovery:** an estimated ₹196,350 recovered under the illustrative ₹50-per-breach assumption. [INSERT VALUE FROM NOTEBOOK: recalculate using Delhivery's actual per-breach cost figure to obtain a production-ready estimate.]
- **Conservative estimate:** this simulation only models the direct effect on segments originating from the three upgraded hubs. It does not capture downstream ripple effects through the network — given that two of these three hubs rank in the top 3 for betweenness centrality, the true network-wide impact of these upgrades is likely materially higher than this direct estimate.

Prioritization Roadmap

Horizon	Action	Rationale
Immediate (0-3 months)	Capacity & process review at Gurgaon Bilaspur HB and Bangalore Nelamangala H	Highest structural risk scores and largest SLA-breach contributions; highest-leverage interventions
Immediate (0-3 months)	Deploy graph-enhanced ETA model to production	Drop-in replacement for existing ML pipeline; 19.1% MAE improvement with no operational change required
Medium-term (3-6 months)	Regional cluster review: Bhiwandi Mankoli HB + Mumbai/Pune corridors; Kolkata Dankuni HB + Eastern feeder corridors	Addresses corridor-level delay clusters identified independently of hub-level centrality
Medium-term (3-6 months)	Re-assign route type on the 3 highest-impact flagged corridors (Bangalore Nelamangala <-	Captures ~46 of the 53 hours of total FTL/Carting recoverable opportunity

	> Bengaluru Bommasandra, Delhi Gateway - > Gurgaon Bilaspur)	
Long-term (6-12 months)	OSRM map-data correction audit for extreme-outlier rural/regional corridors	Improves both customer-facing ETA accuracy and model input quality network-wide
Long-term (6-12 months)	Preventive capacity planning at Hyderabad Shamshabad H ahead of volume growth	Currently structurally important but not yet delay-heavy; lower-cost to address proactively

8. Conclusion

This analysis demonstrates that ETA unreliability at Delhivery is not a diffuse, network-wide problem requiring broad investment — it is concentrated, measurable, and addressable. A small set of structurally critical hubs accounts for a disproportionate share of network-wide delay, a defined set of corridors carries the bulk of recoverable transit time, and a single model upgrade — already validated against historical data — can improve customer-facing ETA accuracy by double digits without any change to the physical network.

We recommend Delhivery move on two parallel tracks: deploy the graph-enhanced ETA model immediately as a low-risk, high-return software upgrade, while launching the prioritized hub and corridor interventions outlined above, beginning with Gurgaon Bilaspur HB and Bangalore Nelamangala H. Together, these actions represent a data-driven, capital-efficient path to materially improving delivery reliability — the foundation of customer trust in Delhivery's network.

We welcome the opportunity to walk through these findings in detail and to support the operational teams in translating this analysis into an executable rollout plan.